Grafica Avansata pe Calculator

Laborator 3
Python si Biblioteci Grafice

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Laborator 3

Unelte necesare:

- **Visual Studio**
- **Python Environments in Visual Studio**
- **Librarii in Python: graphics.py (continuare), Tkinter, PyGame, Turtle, Cgkit**

[7 Best UI Graphics Tools For Python Developers With Starter Codes | by Bharath K | Towards Data Science]

"The Graphics User Interface (GUI) built with Python are extremely useful for a variety of projects. You can use these technologies to make your projects unique, aesthetically pleasing, visually appealing, highly interactive environment, and provide users with other similar wonderful features. You can even use these tools for developing artificial intelligence projects with machine learning or deep learning models to make them stand out from other such project ideas."

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Link-uri utile Python:

- **Introducere in Python**
  www.w3schools.com/python/Python Environments in Visual Studio

- Quick Start Guide for Python in VS Code (visualstudio.com)
  Cum sa creezi si sa deschizi un proiect/fisier cu Python (Engleza)

- Introducere_Python.pdf (uab.ro) dupa Prof. Adriana Birlutiu (Romana)
- Introducere-in-python.pdf (fmi-unibuc-ia.github.io) (Romana)
- CAP (ubbcluj.ro) (Romana)

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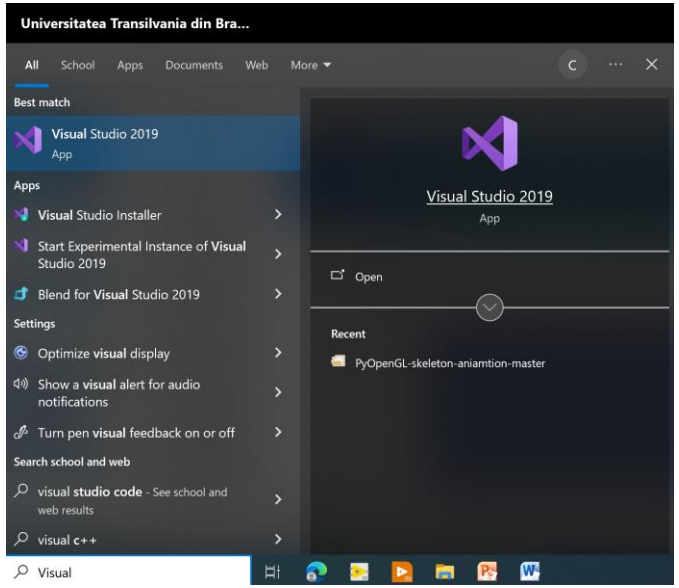
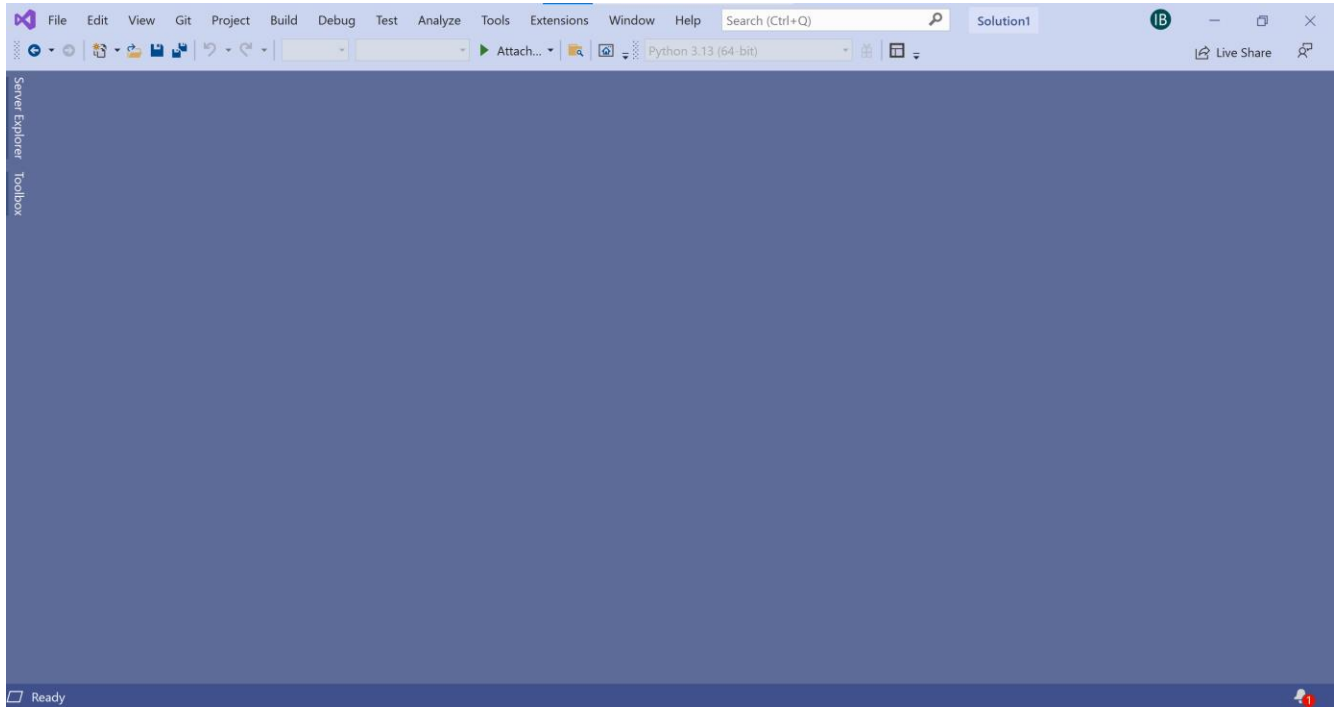
Grafica in Python

Visual Studio

- Deschideti Visual Studio





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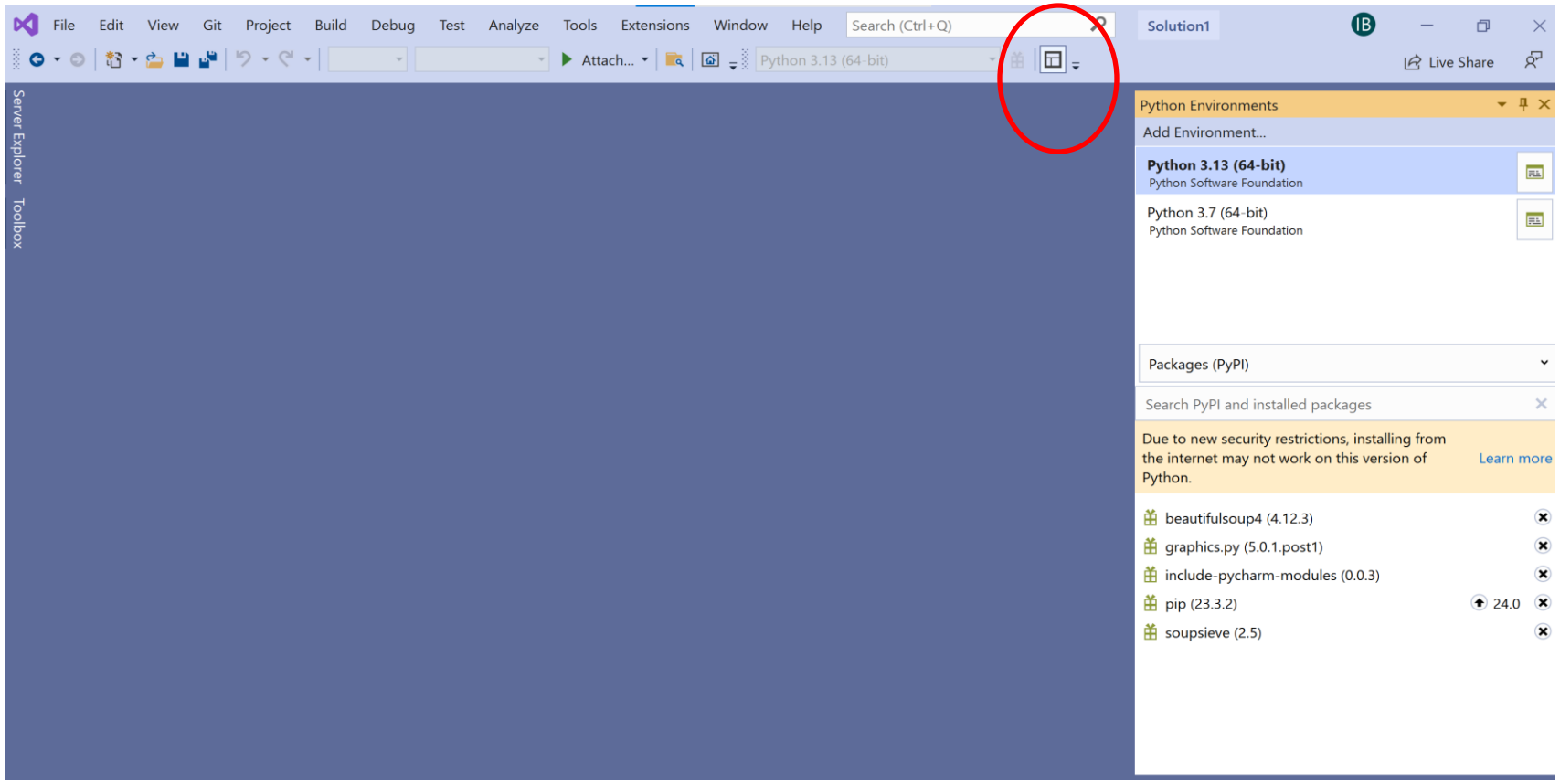
Grafica in Python

Din Visual Studio selectam:

Python Environments in Visual Studio



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Grafica in Python

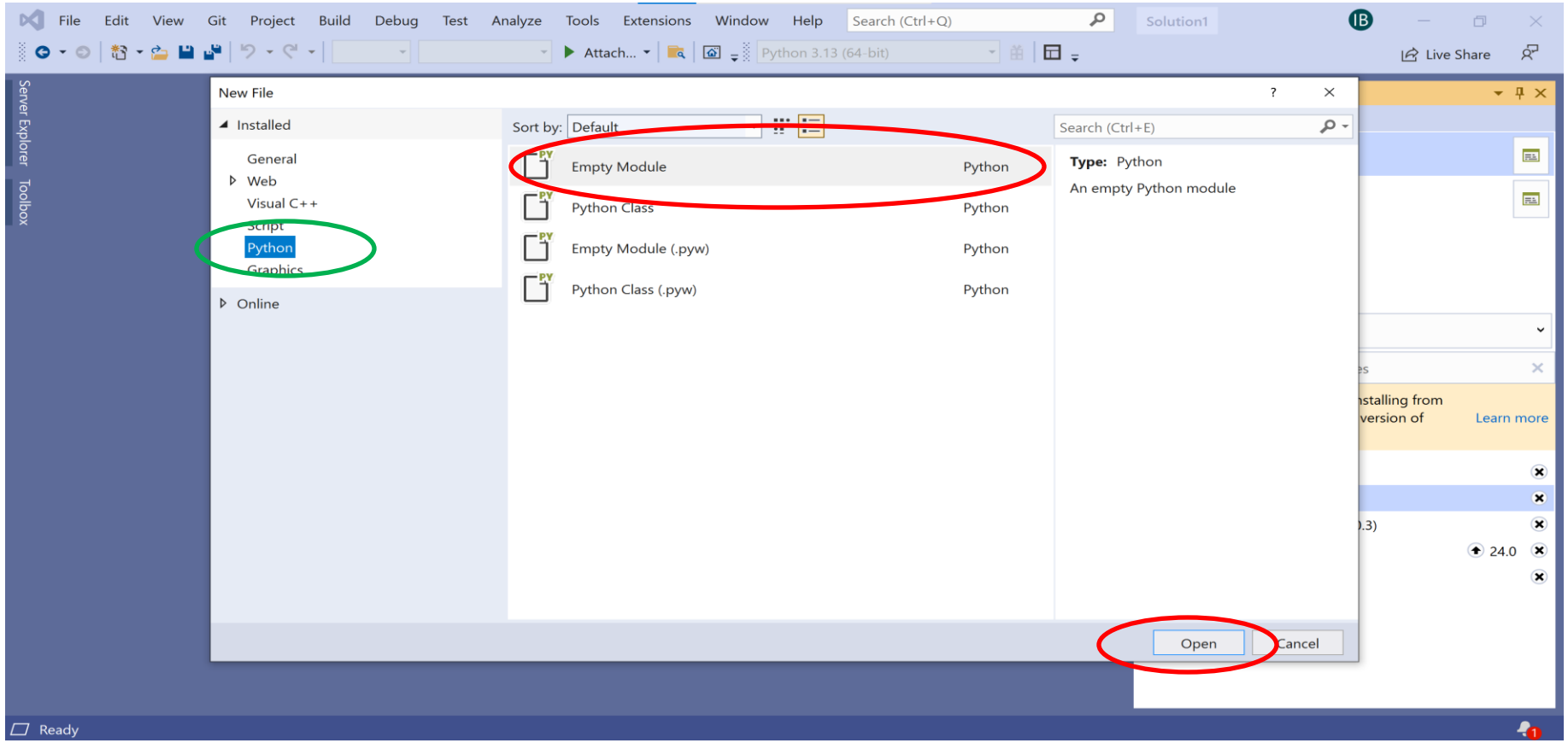
Din Visual Studio

Creaza Program

File ->New->File

Empty Module (Python)

Selecteaza si Valideaza buton **Open**



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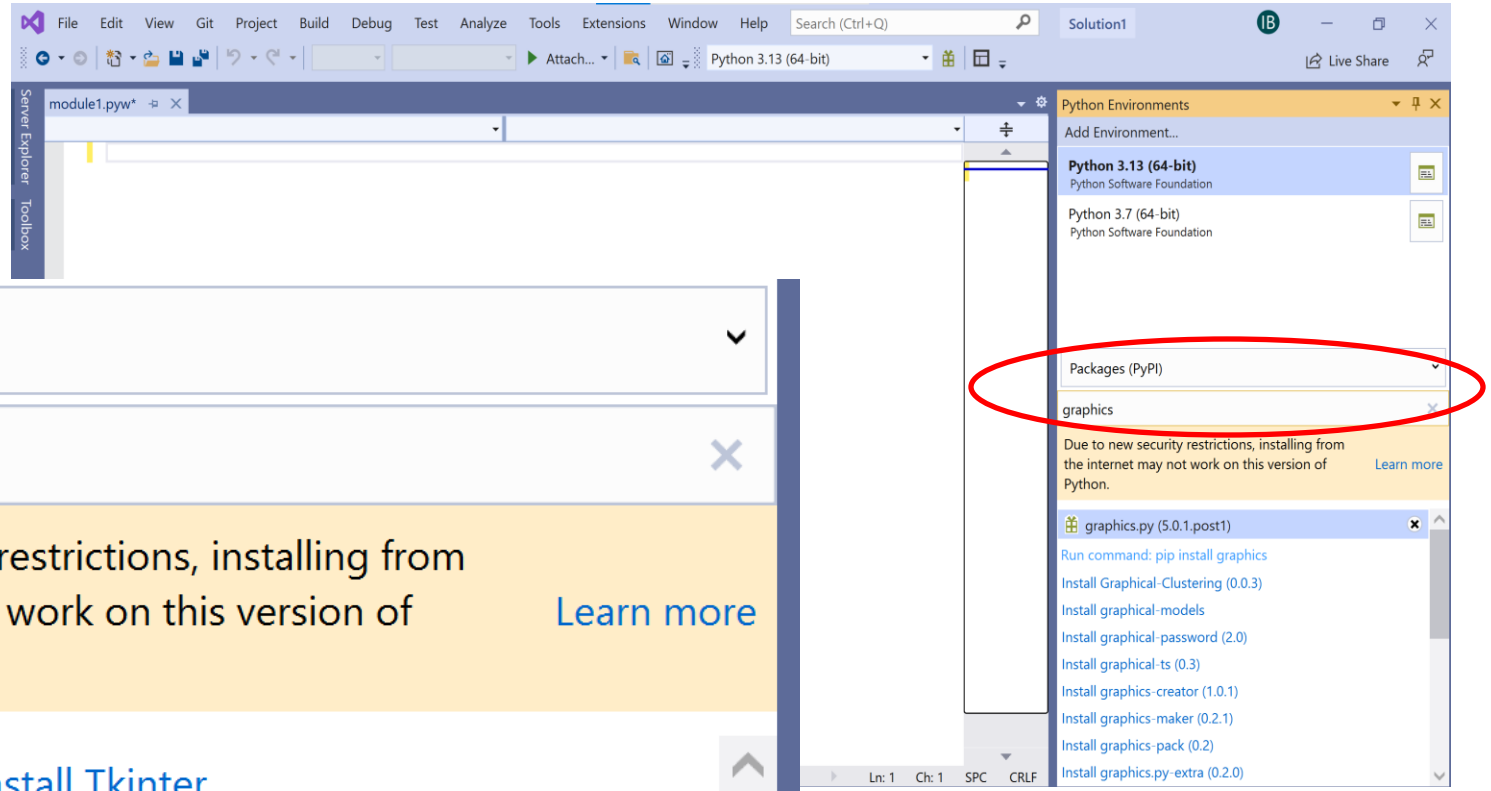
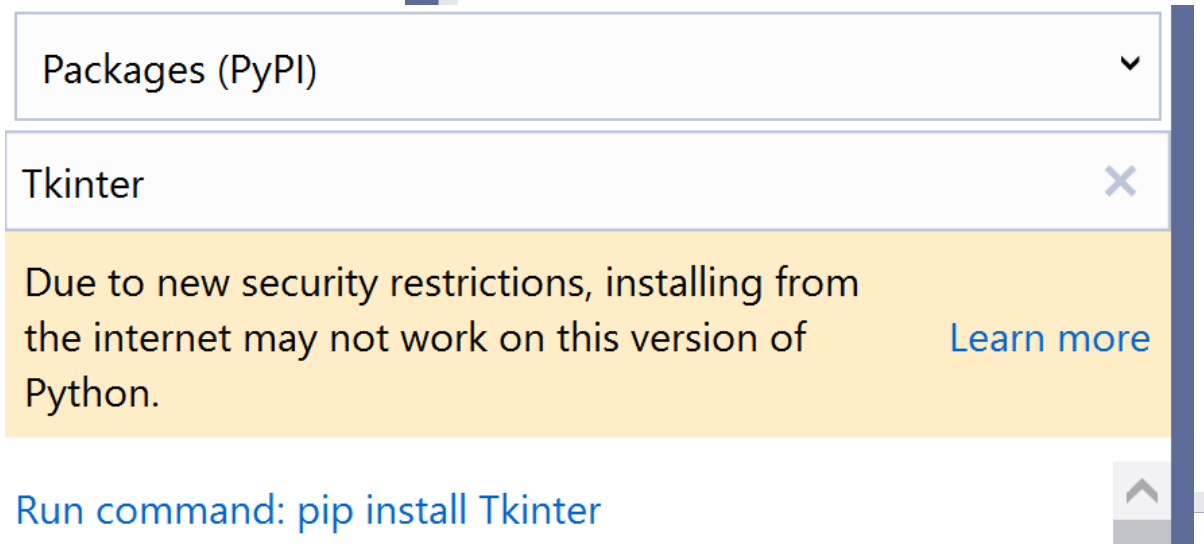
Grafica in Python

Librarii in Python: Tkinter

Instaleaza graphics.py

Instaleaza Tkinter





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Grafica in Python

Librarii in Python: graphics.py, Tkinter

- Instaleaza **graphics.py**,
- Instaleaza **Tkinter**

Pentru a utiliza **GraphWin**, trebuie să descărcați și să importați **graphics.py** și să vă asigurați că aveți Tkinter.

Pentru GraphWin, pachetul de grafică Python de bază pe care îl folosește este **Tkinter**.

Pentru a crea o **fereastră grafică**, se apeleaza constructorul trecând titlul, lățimea și înălțimea. GraphWin Objects (wartburg.edu)

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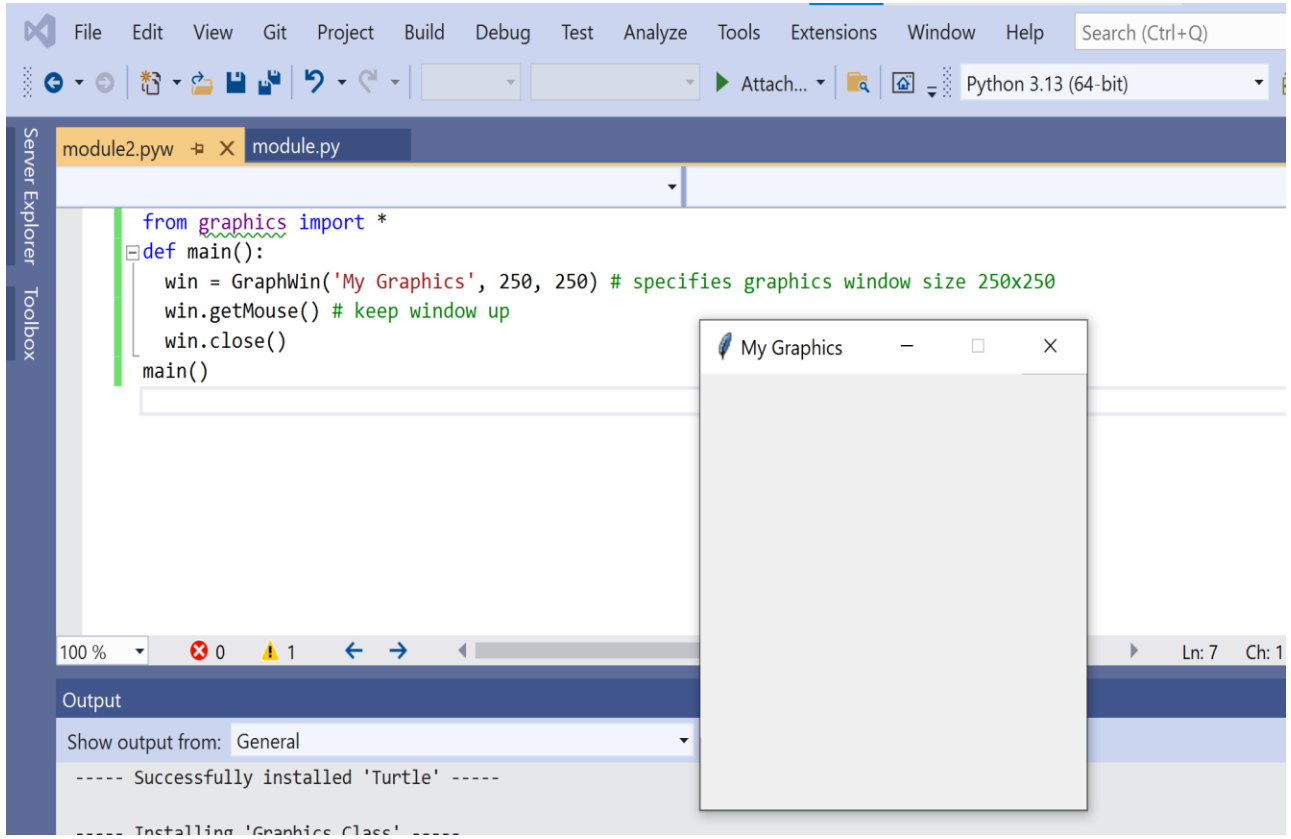
Grafica in Python

Librarii in Python: graphics.py, Tkinter

```
from graphics import *
def main():
  win = GraphWin('My Graphics', 300, 300)
# window size 250x250
  win.getMouse() # pastreaza window up
  win.close()
main()
```



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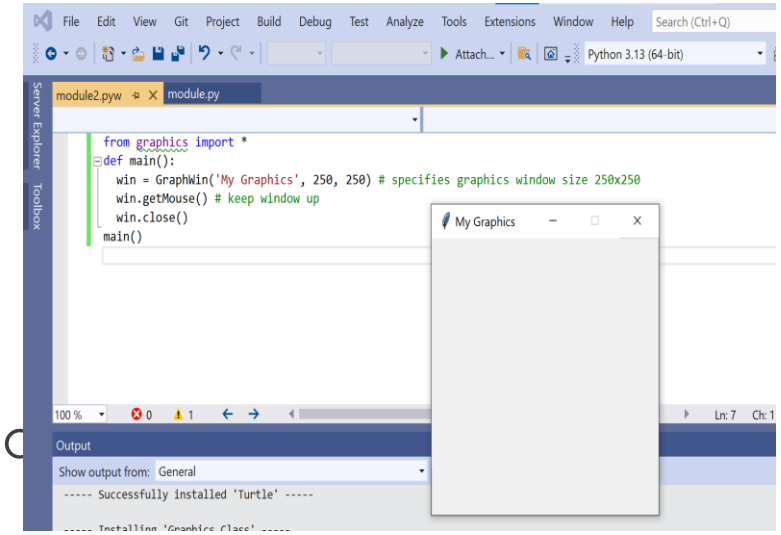
Grafica in Python

Librarii in Python: graphics.py, Tkinter



Task1: Creaza o interfata grafica, caruia ii atribui un nume si o dimensiune 500x500.

(a) Plaseaza pe interfata grafica o imagine gif, caruia sa ii functioneze animatia.

(b) Evaluati ce librarii puteti sa mai folositi pe langa Tkinter. Propuneti variante/solutii.

(c) Salveaza codul si depune-l in raportul final al Lab3.

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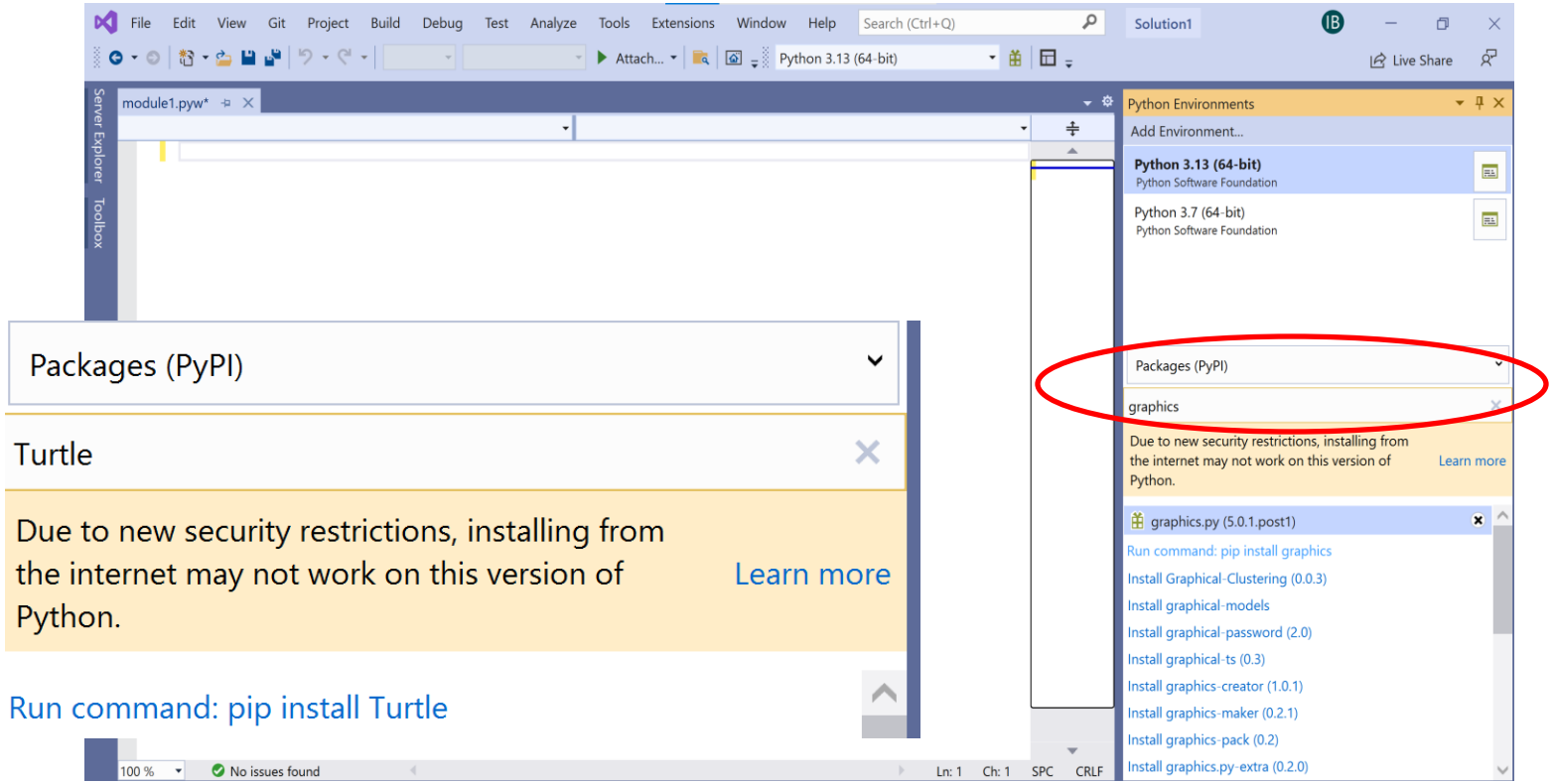
Librarii in Python: Turtle

- Instaleaza graphics.py
- Instaleaza Turtle



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Librarii in Python: Turtle

- turtle — Turtle graphics — Python 3.12.2 documentation
- Python Turtle Programming Tutorial – javatpoint

Task2:

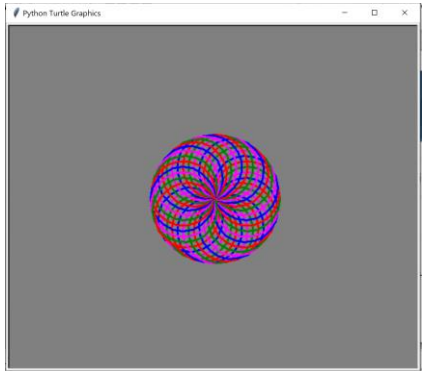
(a) Evaluati scriptul;

(b) Completati la 8 culori pentru generare; schimbati culoare background si viteza de generare.

(c) Realizati un program care sa se genereze dupa un patrat.

(d) Salveaza codul si depune-l in raportul final al Lab3.

Rulati urmatorul script:




```
import turtle
t = turtle.Turtle()
turtle.bgcolor("grey")
turtle.pensize(3)
turtle.speed(1)
while (True):
    for i in range(4):
        for colors in ["red", "blue", "magenta", "green"]:
            turtle.color(colors)
            turtle.circle(50)
            turtle.left(10)
  turtle.hideturtle()
turtle.mainloop()
```

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Librarii in Python: Turtle

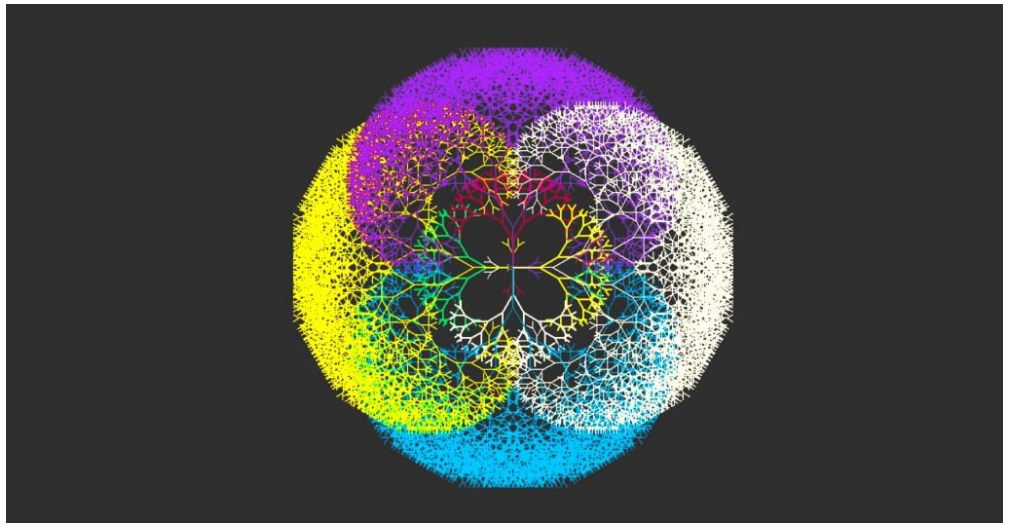
- Turtle Graphics with Python | Aman Kharwal (thecleverprogrammer.com)

Task3:

(a) Evaluati scriptul din link-ul propus mai sus.

(b) Realizati modificari in script pentru a genera o noua imagine fractala.

(c) Salveaza codul si depune-l in raportul final al Lab3.



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Grafica in Python

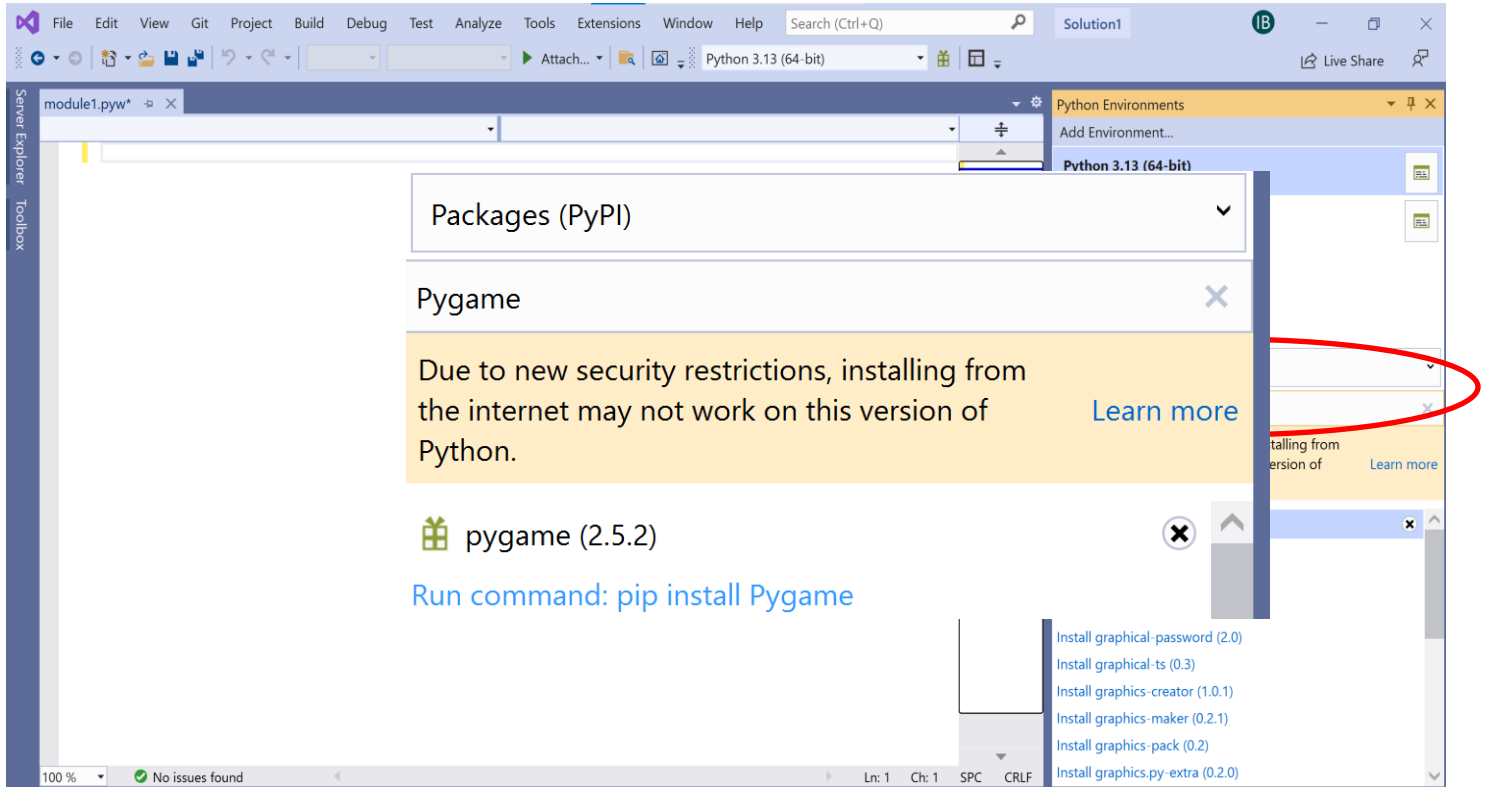
Alte librarii in Python utile: PyGame, Cgkit

- Instaleaza PyGame

[PyGame Tutorial – GeeksforGeeks]

- Instaleaza Cgkit

[1. Introduction — Python Computer Graphics Kit 2.0.0 documentation (sourceforge.net)]



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Grafica in Python; Librarii in Python

1. Pentru cele trei Task-uri, salveaza fisiere in arhiva zip de forma: Grupa_Lab3_Nume_Prenume

2. Pentru fiecare cod comenteaza liniile de cod, definirea punctului, cercului (cercurilor), patratului (patratelor), functiile folosite, structura programului.

3. Realizati o interfata grafica care exprima expresii faciale, miscari, actiune folosindu-va de imagini Gifs. Puteti insera butoane de pornire/oprire Gifs, puteti completa cu generare de figuri geometrice sau imagini fractale.